

FORM TP 2013180



TEST CODE **02220020**

MAY/JUNE 2013

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®
ENVIRONMENTAL SCIENCE
AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

UNIT 2 – Paper 02

2 hours 30 minutes

14 MAY 2013 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Do not open this examination paper until instructed to do so.
2. This paper consists of **SIX** questions, **TWO** from each Module.
3. Answer **ALL** questions.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.



MODULE 1

Answer BOTH questions.

1. (a) Figure 1 shows the results of a survey conducted in two Caribbean countries to determine the public's perception of the level of importance of different threats to sustainable agriculture.

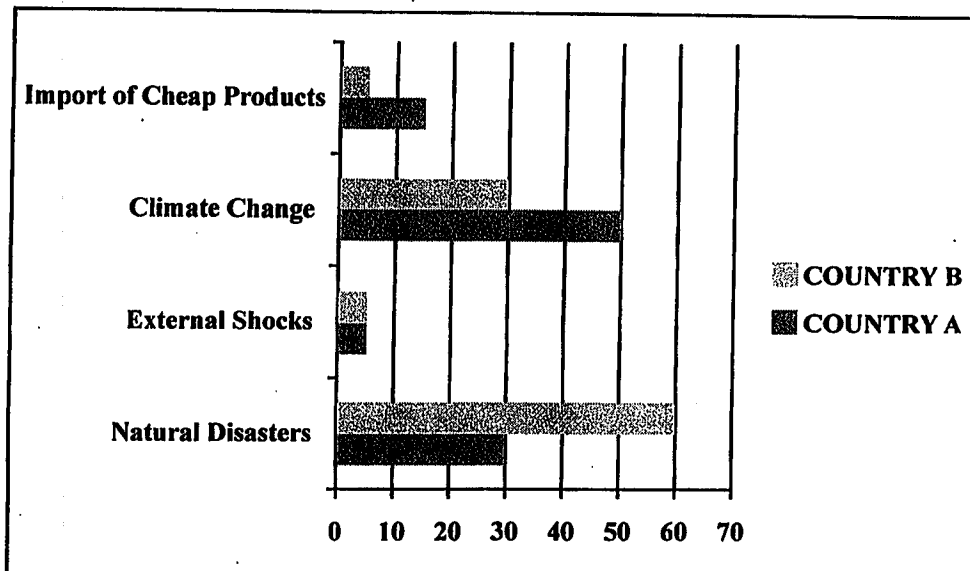


Figure 1. Results of survey to determine the public's perception of the level of importance of different threats to sustainable agriculture

- (i) State THREE features of sustainable agriculture. [3 marks]
- (ii) Using the data in Figure 1, make FIVE deductions about the perception of the level of importance of the different threats to sustainable agriculture. [5 marks]
- (iii) Explain, giving THREE reasons, why 'external shocks' are threats to sustainable agriculture. [6 marks]
- (b) "More Caribbean people should become involved in agriculture because it can contribute to the economies of Caribbean countries". Give TWO reasons to justify this statement. [6 marks]

Total 20 marks

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2. (a) Figure 2 shows the amount of fish produced from traditional fishing grounds and aquaculture over a ten-year period.

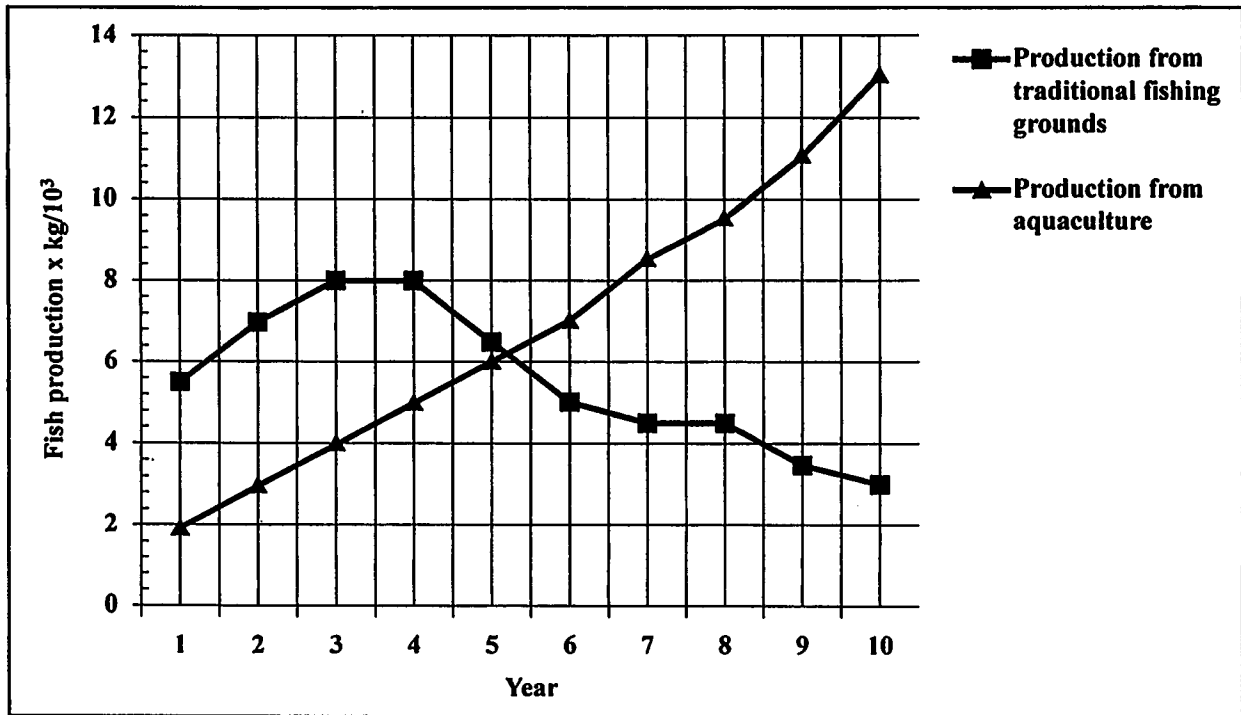


Figure 2. Fish production from traditional fishing grounds and aquaculture

- (i) Describe the trends in fish production illustrated in Figure 2. [5 marks]
- (ii) Suggest TWO reasons why fish production from aquaculture shows the trend illustrated in Figure 2. [4 marks]
- (iii) Outline TWO impacts that this trend in fish production from aquaculture can have on the environment. [4 marks]
- (b) Figure 3 shows the terrain over which a farmer plans to establish a new commercial farm to cultivate crops.



Figure 3. Terrain for establishing a commercial farm

- (i) State THREE characteristics of commercial agriculture systems. [3 marks]
- (ii) Recommend TWO sustainable farming practices that the farmer should use to successfully farm the area. Give ONE reason why EACH practice is recommended. [4 marks]

Total 20 marks

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MODULE 2

Answer BOTH questions.

3. (a) A diagram of a proton exchange membrane fuel cell is provided in Figure 4 below. Study the diagram and identify the substances present at points A, B and C. [3 marks]

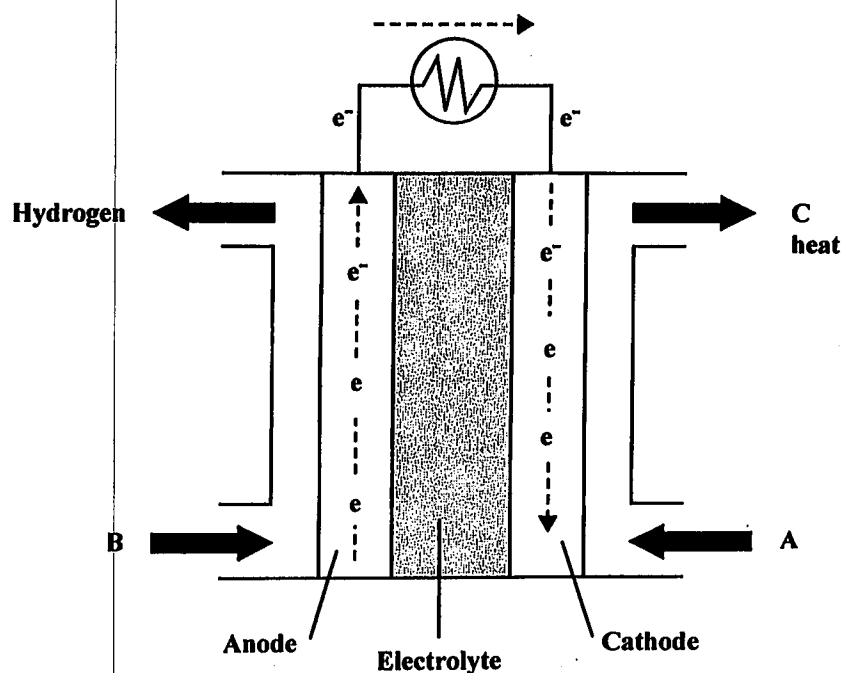


Figure 4. Diagram of a proton exchange fuel cell

- (b) Why is a fuel cell considered a 'secondary energy source'? [3 marks]
- (c) Fuel cells are being considered as a contender for replacing gasoline as the fuel for motor vehicles. Discuss the feasibility of this in the Caribbean, with specific reference to the reliability of supply, and economic factors. [6 marks]
- (d) Outline the production and transmission of electricity using the combustion of a fossil fuel. [3 marks]

- (e) Table 1 below shows the electricity rates in Country X from 1971 to 1990.

TABLE 1: ELECTRICITY RATES IN COUNTRY X FROM 1971 TO 1990

Year	Rate, \$/kWh	Year	Rate, \$/kWh
1971	0.10	1981	0.30
1972	0.10	1982	0.30
1973	0.10	1983	0.30
1974	0.20	1984	0.30
1975	0.20	1985	0.20
1976	0.20	1986	0.30
1977	0.20	1987	0.20
1978	0.20	1988	0.30
1979	0.20	1989	0.30
1980	0.30	1990	0.30

- (i) In your answer booklet, sketch a graph of the data shown in Table 1. (You are NOT required to plot a detailed graph.) [3 marks]
- (ii) State TWO major trends that can be deduced from your sketch. [2 marks]

Total 20 marks

4. (a) A student is conducting an experiment to investigate the hypothesis, "Fluorescent light bulbs are more energy efficient than incandescent ones". To do this, the student sets up the apparatus as shown in Figure 5.

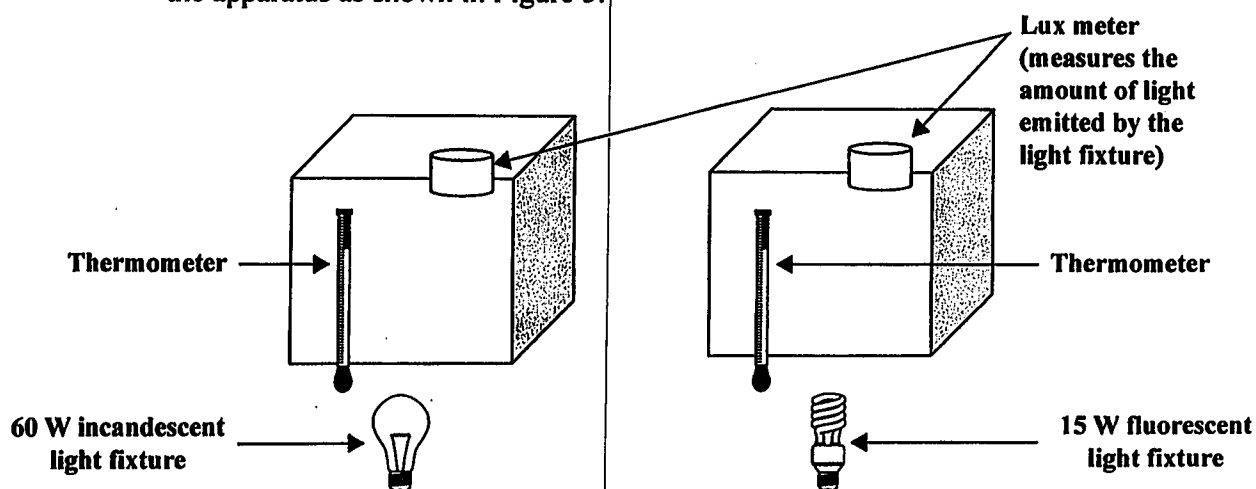


Figure 5. Experimental setup for light bulb efficiency experiment

- (i) Outline THREE steps the student will have to take to conduct this experiment. [3 marks]
- (ii) State THREE precautions the student will have to take to get the most accurate results. [3 marks]

Table 2 below shows the data collected by the student during the experiment.

Table 2: RESULTS OF THE LIGHT BULB EFFICIENCY EXPERIMENT

Type of Light Bulb	Temperature, °C		Amount of Light, Lux
	Before Experiment	After Experiment	
60 W incandescent	28.1	86.7	50
15 W fluorescent	28.1	32.6	50

- (iii) Identify which light bulb is more efficient AND clearly explain how the experimental results allow you to reach your conclusion. [5 marks]
- (b) The total cost of generating electricity by natural gas combustion can be divided into generation, environmental and 'other costs'.
- (i) List THREE costs, other than generating and environmental costs, that could be included in the 'other costs' category. [3 marks]
- (ii) Explain why TWO of the costs listed in Part (b) (i) above should be included in any discussion of the overall cost of using natural gas to produce electricity. [6 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions.

5. (a) Explain the difference between the terms 'pollutant' and 'pollution'. [2 marks]
- (b) Figure 6 illustrates the trends in population numbers and per capita gross national income for a Caribbean country from 1990 to 2005.

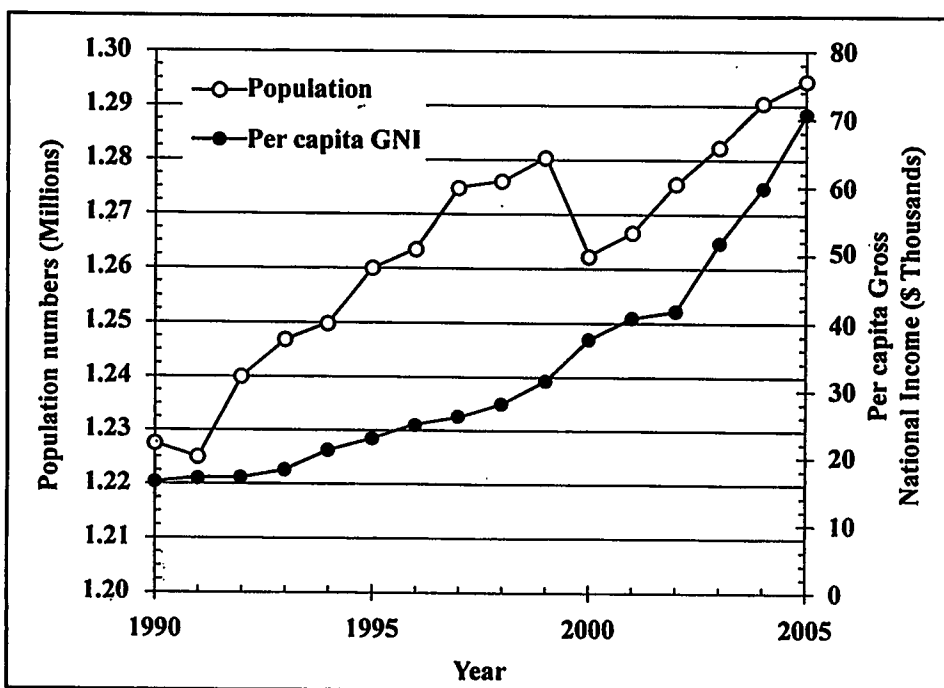


Figure 6. Data on the population and per capita gross national income for a Caribbean country from 1990 to 2005.

- (i) What were the overall rates of increase in the per capita gross national income over the period 1990 – 2000 and 2000 – 2005? [5 marks]
- (ii) The country has recently completed a 'pollution index' exercise for the entire country. The index ranges from 0 to 100 and can be used to rate the level of pollution in the country. A pollution index of 0 indicates no pollution, while 100 indicates a highly polluted environment. In 1990, the pollution index was 14, while in 2005, the pollution index rose to 73. Use the information provided in Figure 6 and your answer to (b) (i) to discuss TWO likely causes of the rise in pollution. [6 marks]
- (c) The Montreal Protocol is an international convention that has proven to be very successful in controlling a certain type of pollution around the world.
- (i) State the name of the group of pollutants that the Montreal Protocol is used to control. [1 mark]
- (ii) Describe the way in which the Montreal Protocol works to control the type of pollution identified in (c) (i) AND provide ONE reason for its success. [6 marks]

Total 20 marks

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6. Figure 7 illustrates the area surrounding Green Valley Village. The residents of the village all receive tap water from the nearby drinking water treatment plant, and also use the river water directly for farming and animal husbandry. The residents have recently been complaining of an unpleasant taste in their tap water, and some farmers are noticing a decline in their crops irrigated by river water. Only Farm A has been unaffected by crop decline. You have been assigned by the Ministry of the Environment to investigate the complaints.

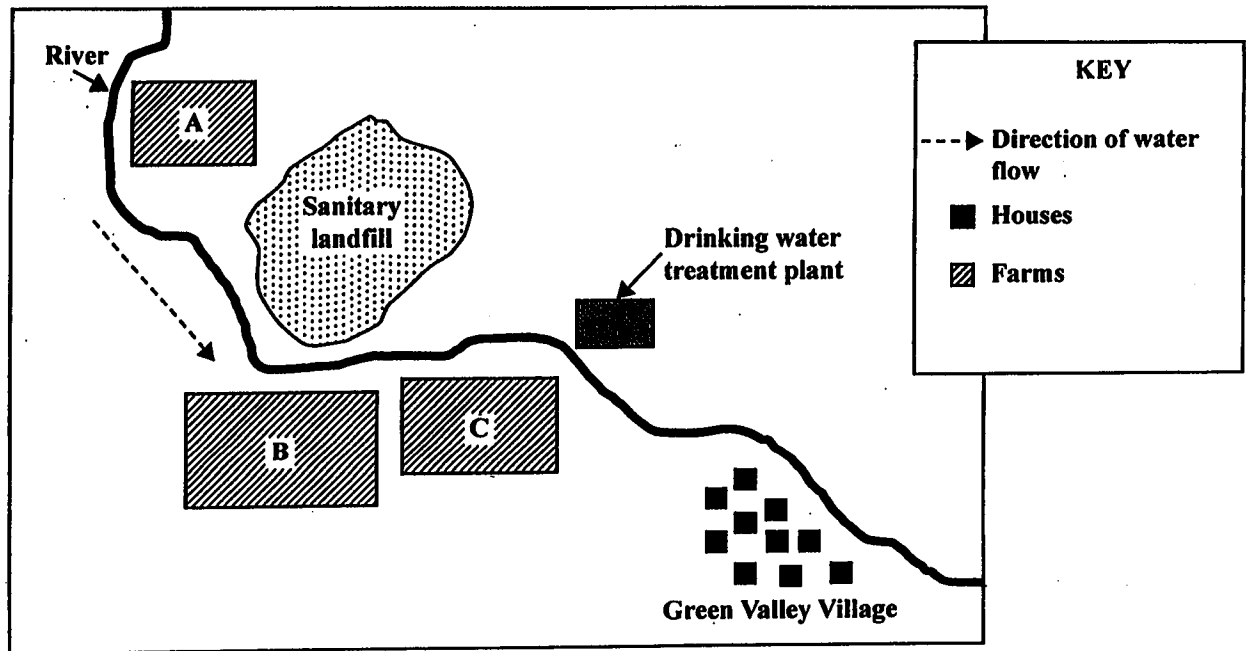


Figure 7. Sketch map of the area surrounding a sanitary landfill

- (a) Outline a plan that will allow you to determine the source of the problems being experienced by the residents and farmers of Green Valley Village. In your response, explain clearly how the plan you propose will allow you to determine the source of the problems. [5 marks]
- (b) State THREE pollutants that are likely to be found in the Green Valley River. [3 marks]
- (c) Persons fishing in the river below Green Valley Village have found that the fishes are highly contaminated with materials they believe came from the landfill. Explain how fish found far from the landfill could be so highly contaminated. [4 marks]
- (d) Suggest TWO measures that can be implemented to mitigate the impact of pollution on the residents and farmers in the Green Valley area. [2 marks]
- (e) The landfill is also a significant source of methane, a greenhouse gas. Explain how methane from the landfill can contribute to global warming. [6 marks]

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.